Sara Tabikh

https://saratabikh.github.io/portfolio/

+61 437 075 001

SARATABIKH1@GMAIL.COM

MELBOURNE, AUSTRALIA

FINAL YEAR AEROSPACE ENGINEERING AND SCIENCE STUDENT

Profile

Driven final-year Aerospace Engineering and Science student with a strong foundation in design, manufacturing, and systems safety. Actively advancing a career in aerospace and defence, supported by hands-on leadership as Safety Lead of a student rocketry team and ongoing professional experience within the aerospace industry.

Professional Experience

BAE Systems Australia — November 2024 – Present

- **Aerospace Engineering Undergraduate.** Developed a program's inaugural Unified Architecture Framework (UAF) model for a System of Systems in Cameo Systems Modeller, collaborating with subject matter experts.
- Improve automation of service desk request handling through the configuration of Jira back-end workflows.
- Conduct User Acceptance Testing in IBM Engineering Test Management.
- Improved internal knowledge management by cataloguing and creating SharePoint sites, collaborating with leadership to refine content for clarity and accessibility.

- **Aerospace Engineering Intern.** Authored full validation report for internal structural analysis tool, enabling automated calculations for aircraft structures team to improve speed and accuracy for high-volume tasks.
- Set up FEA in Patran for aircraft components using original drawings, non-conformance data, and non-destructive testing reports.
- Heavily modified legacy MATLAB code to suit a different vehicle and extracted relevant data.

Monash High Powered Rocketry, Clayton — March 2024 – Present

- **Safety Lead.** Managed a team of 15 people and had a focus on upskilling the entire team (150+ people).
- Revamped legacy safety procedures by standardising Risk Assessments and Safe Work Instructions, successfully integrating safety gates into design reviews to foster a proactive, team-wide safety culture.
- Organised logistics for testing, outreaches and launch events by creating event risk management plans and overseeing a large team to ensure the safety of all attendees.

- **Propulsion Engineer.** Contributed to the design, machining, and testing of a 10 kN liquid oxygen hybrid rocket engine. Involved in testing, static fires, and launches of a 4 kN nitrous oxide hybrid engine, supporting pad setup, engine assembly, flightline and control station operations.
- Developed a resonator system to mitigate combustion instabilities, including Helmholtz resonators, injector-face baffles, and fuel inserts. Completed literature review, PDR/CDR cycles for internal and external stakeholders.
- Designed and integrated components in Creo and Windchill; prepared technical reports and engineering drawings.
- Technical team member competing at both the International Rocket Engineering Competition (IREC) and UK propulsion competition, Race 2 Space, in 2025.

- **Safety Representative.** Alongside Propulsion Engineer responsibilities, enabled safe testing of niche, high-risk propulsion systems, including high-voltage spark ignitors, by developing thorough risk assessments and safe work instructions written in collaboration with part designers and academic advisors.
- Contributed to safety and process improvements, which were part of the work recognised with the Jim Furfaro Award for Technical Excellence among 140+ student teams internationally.

Monash Motorsport, Clayton March 2022 – December 2023

- **Aerodynamics Engineer.** Worked in a small team to design a high downforce and lightweight rear wing whilst meeting Formula SAE rules using Siemens NX, SOLIDWORKS and Ansys Fluent.
- Fabricated 15+ complex composite structures (aerodynamic, load-bearing, and sandwich panels) via wet-lay and infusion using Kevlar and diverse carbon fibre weaves.
- Worked on validating CFD analysis of the vehicle by comparing simulations to data from wind tunnel testing at Monash University. Utilised pressure tapped aerodynamic devices and pressure rakes to understand the pressure distribution near aerodynamic devices.

- **Vehicle Dynamics Engineer.** Used vehicle simulations to predict and enhance vehicle performance and lap times. Identified effects of set-up changes and testing conditions through data analysis.
- Worked efficiently under pressure during vehicle testing and annual Formula SAE Competition events and decisively responded to unexpected challenges faced.

Resonance in Supersonic Intakes, Shock Lab Research Group February 2026 – Present

- Individual Final Year Project involving CFD, design, testing and analysis of intakes using a supersonic wind tunnel and high speed schlieren imaging.
- Investigating types of resonance found in supersonic inlets at overspeed mode, known as the buzz mechanism.

Supervisor, Only Hospitality April 2021 – November 2024

- Leadership responsibilities including opening and closing the store independently and providing team leadership throughout the day.

Certificates and Awards

Jim Furfaro Award for Technical Excellence, International Rocketry Engineering Competition 2025 June 2025

MSC NASTRAN – Linear Static & Normal Modes Analysis & PATRAN – Introduction, Hexagon Manufacturing Intelligence December 2024 – January 2025

Monash Machining Industry Skills Course, Chisholm Institute June 2023 – August 2023

SolidWorks Mechanical Design, Monash University May 2023

Introduction to ANSYS FEA and CFD, Leap Australia April 2023

Skills

Software: SOLIDWORKS, PTC Creo, PTC Windchill, Siemens NX, Cameo Systems Modeler, Jira, IBM Jazz Toolsuite (DOORS Next, Engineering Workflow Management, Engineering Test Management)

Manufacturing: Composite manufacturing, CNC & manual machining (lathe and mill)

Simulation and Analysis: Ansys Computational Fluid Dynamics (CFD), Finite Element Analysis (FEA), MSC NASTRAN, PATRAN, Abaqus (composite FEA)

Languages: MATLAB, Python, Java

Education

Monash University, Bachelor of Engineering (Honours) and Science Expected 2026

Specialisation: Aerospace Engineering, Major: Astrophysics